The Fabric ID is a 64-bit value that identifies the Fabric and is scoped to a particular Root CA. For example, two fabrics with the same Fabric ID are not equivalent unless their Root CA are the same. The Fabric ID MAY be chosen randomly or algorithmically but it SHALL be allocated uniquely within the set of all possible Fabric IDs for which a given Root CA will sign operational certificates. Before allocating the Fabric ID, the Commissioner SHOULD attempt to ensure that an existing Fabric is reused and joined, if any is applicable from the perspective of the Commissioner in the current commissioning context. The method used for determining local Fabric ID existence is vendor-specific.

The Node ID is a 64-bit value that identifies a Node within a Fabric. The Node ID MAY be chosen randomly or algorithmically but it SHALL be allocated uniquely within the Fabric before it is given to the Node or otherwise used. The Node ID SHALL be chosen, by a Commissioner, at the time of Node commissioning.

The uniqueness constraint for Fabric ID is only required to be ensured within the scope of the Root CA serving the Commissioner.

6.4.4. Node Operational Key Pair

A Node Operational Key Pair, comprised of a Node Operational Public Key and a Node Operational Private Key, is created using the `Crypto_GenerateKeypair` function. A new Node Operational Key Pair is generated for each Commissioning Session in accordance with security requirements.

6.4.5. Node Operational Credentials Certificates

All certificates in the Node Operational credentials are X.509v3 certificates compliant with RFC 5280, encoded in such a way that they respect the constraints in the Operational_Certificate section. They may be encoded as X.509v3 certificates or Matter Operational Certificates ("Matter Certificates" thereafter). The signature field of a certificate SHALL be calculated using the X.509v3 encoding of the certificate.

6.4.5.1. Node Operational Certificate (NOC)

The NOC SHALL be issued by either a Root CA trusted within the Fabric or by an Intermediate Certificate Authority (ICA) whose ICA certificate is directly issued by such a Root CA. The NOC is bound to the Node Operational Key Pair through the Node Operational Credential Signing Request (NOCSR).

The validity period specifies the time period for which a NOC is valid. For constrained or sleepy devices that lack accurate time, enforcement of an NOC’s validity period MAY be omitted.

6.4.5.2. Intermediate CA (ICA) Certificate

In the case where an intermediate CA (ICA) issues the NOC, the ICA certificate is used to attest to the validity of the NOC. The Root CA certificate associated with the issuer of the ICA certificate is used in turn to attest to the validity of the ICA certificate.